Conditional Statement

Session Objectives

- Understand what control statements are.
- Understand the importance of control statements.
- Understand the types of control statements.
- Understand what conditional statements are.
- Understand what nested conditional statements are.
- Understand conditional statements with shorthand notations.

```
# min() , max() and sum()
_dict = {
    ('stud1','stud2') : 98,
    ('stud3','stud4') : 77,
    ('stud5','stud6') : 81,
    ('stud7','stud8') : 91,
    ('stud9','stud10') : 89
}
print(min(_dict.values())) # 77 is min [98,77,81,91,89]
```

```
77
```

```
print(max(_dict.values()))
```

```
98
```

```
print(sum(_dict.values()))
```

```
436
```

```
min(_dict.keys())
```

```
('stud1', 'stud2')
```

```
max(_dict.keys())
```

```
('stud9', 'stud10')
```

```
# min() , max() and sum()
_dict = {
    98 : ['stud1','stud2'],
    77 : ['stud3','stud4'],
    81 : ['stud5','stud6'],
    91 : ['stud7','stud8'],
    89 : ['stud9','stud10']
}
print(min(_dict.keys())) # 77 is min [98,77,81,91,89]
```

```
77
```

```
print(max(_dict.keys()))
```

```
98
```

```
print(sum(_dict.keys()))
```

```
436
```

```python
# Changing or adding an item in Dictionary
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
student_details['city'] = "Ajmer"
```

```python
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Ajmer',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP']}
```

```python
student_details['email'] = 'siddharth.sharan@gmail.com'
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Ajmer',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP'],
 'email': 'siddharth.sharan@gmail.com'}
```

```python
# .update()
stud_info = {'age' : 30 , 'city' : 'Jaisalmeer' , 'course':'Data Science with GenAI'}
student_details.update(stud_info)
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 30,
 'gender': 'Male',
 'city': 'Jaisalmeer',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science with GenAI',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP'],
 'email': 'siddharth.sharan@gmail.com'}
```

```python
# .update()
stud_info = {'age' : 29 , 'city' : 'Udaipur' , 'course':'Data Science with NLP',
             'phone_number' : '9876543210'}
student_details.update(stud_info)
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Udaipur',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science with NLP',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP'],
 'email': 'siddharth.sharan@gmail.com',
 'phone_number': '9876543210'}
```

```python
# Zip Function -> combine 2 iterables into key-value pairs
key = ['name','age','gender','city','state','country']
value = ('Pratyush',27,'Male','Lucknow','Uttar Pradesh','India')
zip_dict = zip(key,value)
zip_dict
```

```
<zip at 0x29b22614140>
```

```python
dict(zip_dict)
```

```
{'name': 'Pratyush',
 'age': 27,
 'gender': 'Male',
 'city': 'Lucknow',
 'state': 'Uttar Pradesh',
 'country': 'India'}
```

```python
list(zip_dict)
```

```
[]
```

```python
# Zip Function -> combine 2 iterables into key-value pairs
key = ['name','age','gender','city','state','country']
value = ('Pratyush',27,'Male','Lucknow','Uttar Pradesh','India')
zip_dict = zip(key,value)
zip_dict
```

```
<zip at 0x29b22600900>
```

```python
list(zip_dict)
```

```
[('name', 'Pratyush'),
 ('age', 27),
 ('gender', 'Male'),
 ('city', 'Lucknow'),
 ('state', 'Uttar Pradesh'),
 ('country', 'India')]
```

```python
tuple(zip_dict)
```

```
()
```

```python
# Removing the Dictionary Items
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
del student_details['country']
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Jaipur',
 'state': 'Rajasthan',
 'course': 'Data Science',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP']}
```

```python
# Not Providing the Key while Deleting : Then it will delete the complete Dictionary
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
del student_details # delete the entire Dictionary
```

```python
print(student_details) # NameError
# NameError: name 'student_details' is not defined
```

```python
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
del student_details['skills'][-1] # ['NLP']
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Jaipur',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science',
 'skills': ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL']}
```

```python
# .pop() -> Removes a specific key and returns it value
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
skills_info = student_details.pop('skills')
skills_info
```

```
['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP']
```

```python
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Jaipur',
 'state': 'Rajasthan',
 'country': 'India',
 'course': 'Data Science'}
```

```python
# .popitem() -> Removes the Last Inserted Values
course_details = student_details.popitem()
print(course_details)
```

```
('course', 'Data Science')
```

```python
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Jaipur',
 'state': 'Rajasthan',
 'country': 'India'}
```

```python
country_detail = student_details.popitem()
print(country_detail)
```

```
('country', 'India')
```

```python
student_details
```

```
{'name': 'Siddharth Sharan',
 'age': 29,
 'gender': 'Male',
 'city': 'Jaipur',
 'state': 'Rajasthan'}
```

```python
# Shallow Copy() -> Creating an independent Object in 2 Different Address.
# Changing on one wouldn't impact other.
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
stud_details = student_details.copy()
print(id(stud_details))
print(id(student_details))
```

```
2865320437056
2865320436480
```

```python
# Shallow Copy -> dict() Dictionary as Constructor
shallow_dict = dict(stud_details)
print(id(stud_details))
print(id(student_details))
print(id(shallow_dict))
```

```
2865320437056
2865320436480
```

Conditional Statement ¶

```python
# If condition:
    # code block
val = int(input("Enter the Value: "))
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")

print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  121
121 is greater than 100.
Outside the If Condition, It Always Works!
```

```python
val = int(input("Enter the Value: "))
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")

print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  77
Outside the If Condition, It Always Works!
```

Control statements in Python are like traffic signals for your code — they decide which parts of your program should run and when.

Traffic light analogy:

Green → Go → Run code block

Red → Stop → Skip code block

Syntax : If

```
If condition:
    # code block
```

Syntax : If-else

```
if condition:
    # Code block to execute if the condition is true
else:
    # Code block to execute if the condition is false
```

Syntax : If-elif-else

```
if condition1:
    # Block if condition1 true
elif condition2:
    # Block if condition2 true
elif condition3:
    # Block if condition3 true
...
else:
    # Block if all conditions are false
```

Syntax : Nested-if

```
if condition1:
    # Block if condition1 is true
    if condition2:
        # Block if both condition1 and condition2 are true
```

Syntax : Nested if-else

```
if condition1:
    if condition2:
        # Block if both condition1 & condition2 true
    else:
        # Block if condition1 true, condition2 false
else:
    # Block if condition1 is false

or
if condition:
    # Block if condition is true
else:
    if another_condition:
        # Block if another_condition is true
    else:
        # Block if another_condition is false
```

Syntax : Nested if-elif-else

```
if condition1:
    if condition2:
        # Block if both condition1 & condition2 true
    elif condition3:
        # Block if condition1 true, condition2 false, condition3 true
    else:
        # Block if condition1 true, others false
elif condition4:
    # Block if condition1 false, condition4 true
else:
    # Block if all conditions false
```

```
num1 = int(input("Enter the Num1: "))
num2 = int(input("Enter the Num2: "))

if num1>num2:
    print(f"{num1} is greater than {num2}")
if num1<num2:
    print(f"{num1} is less than {num2}")
if num1==num2:
    print(f"{num1} is equal to {num2}")
if num1!=num2:
    print(f"{num1} is not equal to {num2}")
```

```
Enter the Num1:  99
Enter the Num2:  77
99 is greater than 77
99 is not equal to 77
```

```
_bool = True
int(_bool) # 1
```

```
1
```

```
_bool = False
int(_bool) # 0
```

```
0
```

```
# condition True : Comparison Operator , Positive [100,912] & Negative [-11,-77] , Filled Iterable
# condition False : Comparison Operator , 0 , Empty Iterables -> [],{},set(),()
```

```
val = int(input("Enter the Value: "))
if 0: # False
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  111
Outside the If Condition, It Always Works!
```

```
val = int(input("Enter the Value: "))
if (): # False
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  91
Outside the If Condition, It Always Works!
```

```
val = int(input("Enter the Value: "))
if set(): # False
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  81
Outside the If Condition, It Always Works!
```

```python
val = int(input("Enter the Value: "))
if 121: # True
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  77
77 is greater than 100.
Outside the If Condition, It Always Works!
```

```python
val = int(input("Enter the Value: "))
if -11: # True
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  99
99 is greater than 100.
Outside the If Condition, It Always Works!
```

```python
val = int(input("Enter the Value: "))
if (1,2,3,4,5): # True
    print(f"{val} is greater than 100.")
print("Outside the If Condition, It Always Works!")
```

```
Enter the Value:  55
55 is greater than 100.
Outside the If Condition, It Always Works!
```

```python
if condition:
    # Code block to execute if the condition is true
else:
    # Code block to execute if the condition is false
```

```python
val = int(input("Enter the Value: "))
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")
else:
    print(f"{val} is less than or equal to 100.")
```

```
Enter the Value:  121
121 is greater than 100.
```

```python
val = int(input("Enter the Value: "))
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")
else:
    print(f"{val} is less than or equal to 100.")
```

```
Enter the Value:  77
77 is less than or equal to 100.
```

```python
val = int(input("Enter the Value: "))
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")
else:
    print(f"{val} is less than or equal to 100.")
```

```
Enter the Value:  -21
-21 is less than or equal to 100.
```

```python
val = input("Enter the Value: ")
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")
else:
    print(f"{val} is less than or equal to 100.")
```

```
Enter the Value:  a
```

Fix Code

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[51], line 2
      1 val = input("Enter the Value: ")
----> 2 if val > 100: # Boolean Returns
      3     print(f"{val} is greater than 100.")
      4 else:

TypeError: '>' not supported between instances of 'str' and 'int'
```

```python
val = input("Enter the Value: ")
if val > 100: # Boolean Returns
    print(f"{val} is greater than 100.")
else:
    print(f"{val} is less than or equal to 100.")
```

```
Enter the Value:  a
TypeError: '>' not supported between instances of 'str' and 'int'
```

```python
# Checking the multiple of 10
val = int(input("Enter the Value: "))
if val % 10 == 0: # Boolean Returns
    print(f"{val} is a multiple of 10.")
else:
    print(f"{val} is not a multiple of 10.")
```

```
Enter the Value:  700
700 is a multiple of 10.
```

```python
# Checking the multiple of 10
val = int(input("Enter the Value: "))
if val % 10 == 0: # Boolean Returns
    print(f"{val} is a multiple of 10.")
else:
    print(f"{val} is not a multiple of 10.")
```

```
Enter the Value:  911
911 is not a multiple of 10.
```

```python
# Checking the type of an input()
val = "11"
print(type(val)) # 'str'
if val == 11: # Boolean Returns [False]
    print(f"{val} is a numeric value.")
else:
    print("Please fix the data type of the input, as it is not numeric type.")
```

```
<class 'str'>
Please fix the data type of the input, as it is not numeric type.
```

```python
# Checking the type of an input()
val = 11.99
print(type(val)) # 'float'
if type(val) is float: # Boolean Returns [True]
    print(f"{val} is a float value.")
else:
    print("Please fix the data type of the input, as it is not float type.")
```

```
<class 'float'>
11.99 is a float value.
```

```python
# Checking the type of an input()
val = "11"
print(type(val)) # 'str'
if type(val) is float: # Boolean Returns [False]
    print(f"{val} is a float value.")
else:
    print("Please fix the data type of the input, as it is not float type.")
```

```
<class 'str'>
Please fix the data type of the input, as it is not float type.
```

```python
# Checking the type of an input()
val = 10
print(type(val)) # 'int'
if type(val) is int: # Boolean Returns [True]
    print(f"{val} is a Integer value.")
```

```
<class 'int'>
10 is a Integer value.
```

```python
# Checking the type of an input()
val = 3.14
print(type(val)) # 'float'
if isinstance(val, (int,bool,float,str,complex)): # Boolean Returns [True]
    print(f"{val} is a Float value.")
else:
    print('Not in a Instance List.')
```

```
<class 'float'>
3.14 is a Float value.
```

```python
# Checking the type of an input()
val = '11'
print(type(val)) # 'str'
if isinstance(val, (int,bool,float,complex)): # Boolean Returns [False]
    print(f"{val} is a Str input.")
else:
    print(f'{type(val)} is not present in a Instance List.')
```

```
<class 'str'>
<class 'str'> is not present in a Instance List.
```

```python
# 'in'/'not in' -> Membership Operators to validate a List/Tuple storing 'str' type
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car = 'Innova'
if fav_car in car_list: # True
    print(f"{fav_car} is available in our showroom.")
else:
    print(f"Sorry! Your {fav_car} is not available in our showroom. We'll Add it Soon!")
```

```
Innova is available in our showroom.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car = 'Hummer'
if fav_car in car_list: # False
    print(f"{fav_car} is available in our showroom.")
else:
    print(f"Sorry! Your {fav_car} is not available in our showroom. We'll Add it Soon!")
```

```
Sorry! Your Hummer is not available in our showroom. We'll Add it Soon!
```

```python
if condition1:
    # Block if condition1 true
elif condition2:
    # Block if condition2 true
elif condition3:
    # Block if condition3 true
...
else:
    # Block if all conditions are false
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Creta' # True
fav_car2 = 'RangeRover' # False
if fav_car1 in car_list or fav_car2 in car_list:  # True
    print("We are evaluating our stock_list to check if we have both cars available or not.")
    if fav_car1 in car_list: # True
        print(f"{fav_car1} is available in our showroom.")
    if fav_car2 in car_list: # False
        print(f"Sorry! Your {fav_car2} is not available in our showroom. We'll Add it Soon!")
else:
   print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
We are evaluating our stock_list to check if we have both cars available or not.
Creta is available in our showroom.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Hummer' # False
fav_car2 = 'Baleno' # True
if fav_car1 in car_list or fav_car2 in car_list:  # True
    print("We are evaluating our stock_list to check if we have both cars available or not.")
    if fav_car1 in car_list: # False
        print(f"Sorry! Your {fav_car1} is not available in our showroom. We'll Add it Soon!")
    if fav_car2 in car_list: # True
        print(f"{fav_car2} is available in our showroom.")
else:
   print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
We are evaluating our stock_list to check if we have both cars available or not.
Baleno is available in our showroom.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Creta' # True
fav_car2 = 'Baleno' # True
if fav_car1 in car_list or fav_car2 in car_list:  # True
    print("We are evaluating our stock_list to check if we have both cars available or not.")
    if fav_car1 in car_list: # True
        print(f"{fav_car1} is available in our showroom.")
    if fav_car2 in car_list: # True
        print(f"{fav_car2} is available in our showroom.")
else:
   print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
We are evaluating our stock_list to check if we have both cars available or not.
Creta is available in our showroom.
Baleno is available in our showroom.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Kylaq' # False
fav_car2 = 'Virtus' # False
if fav_car1 in car_list or fav_car2 in car_list:  # False
    print("We are evaluating our stock_list to check if we have both cars available or not.")
    if fav_car1 in car_list: # True
        print(f"{fav_car1} is available in our showroom.")
    if fav_car2 in car_list: # True
        print(f"{fav_car2} is available in our showroom.")
else:
   print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Defender' # True
fav_car2 = 'Virtus' # False
if fav_car1 in car_list and fav_car2 in car_list: # False [T,F]
    print("Hey! We have Both cars in our stock.")
elif fav_car1 in car_list: # True
    print(f"Hey! We Found your {fav_car1} in our stock.")
elif fav_car2 in car_list:
    print(f"Hey! We Found your {fav_car2} in our stock.")
else: # [F,F]
    print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
Hey! We Found your Defender in our stock.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'RangeRover' # False
fav_car2 = 'Innova' # True
if fav_car1 in car_list and fav_car2 in car_list: # False [T,F]
    print("Hey! We have Both cars in our stock.")
elif fav_car1 in car_list: # False
    print(f"Hey! We Found your {fav_car1} in our stock.")
elif fav_car2 in car_list: # True
    print(f"Hey! We Found your {fav_car2} in our stock.")
else: # [F,F]
    print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
Hey! We Found your Innova in our stock.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Defender' # True
fav_car2 = 'Thar' # True
if fav_car1 in car_list and fav_car2 in car_list: # True [T,T]
    print("Hey! We have Both cars in our stock.")
elif fav_car1 in car_list:
    print(f"Hey! We Found your {fav_car1} in our stock.")
elif fav_car2 in car_list:
    print(f"Hey! We Found your {fav_car2} in our stock.")
else: # [F,F]
    print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
Hey! We have Both cars in our stock.
```

```python
car_list = ['Thar','Venue','Creta','Innova','Vitara','ScorpioN',' Brezza','Baleno','Jeep Compass','Defender']
fav_car1 = 'Hummer' # False
fav_car2 = 'Virtus' # False
if fav_car1 in car_list and fav_car2 in car_list: # False [F,F]
    print("Hey! We have Both cars in our stock.")
elif fav_car1 in car_list: # False
    print(f"Hey! We Found your {fav_car1} in our stock.")
elif fav_car2 in car_list: # False
    print(f"Hey! We Found your {fav_car2} in our stock.")
else: # Automatically True as above all are False
    print(f"Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.")
```

```
Sorry, for the Inconvenience, At this moment we don't have both cars that you are looking for.
```